

UPEN

**UNIVERSIDAD POLITÉCNICA
DEL ESTADO DE NAYARIT**



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PORTAFOLIO NUMERO UNO

1. Class with attributes and methods

Cellphone class:

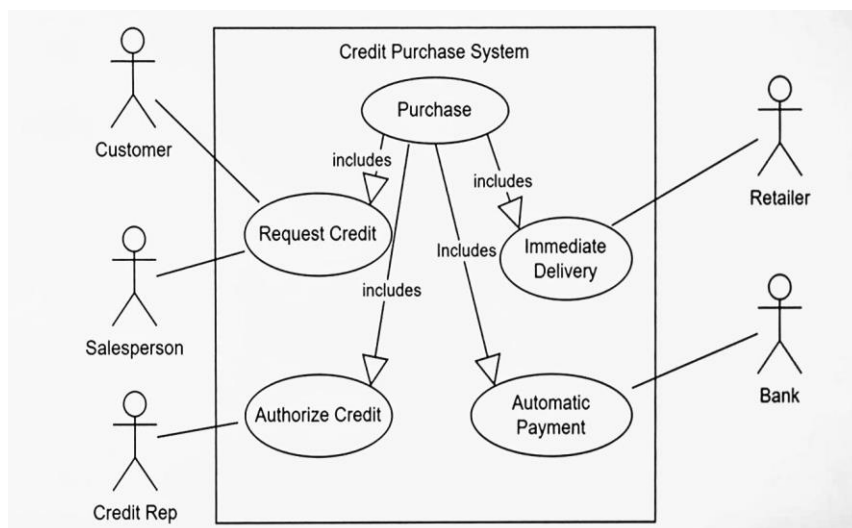
ID: 0123456789

- Size: 6.67"
- Color: White
- Brand: POCO
- Model: F6 Pro
- Phone Number: 311 108 72 97

Main methods:

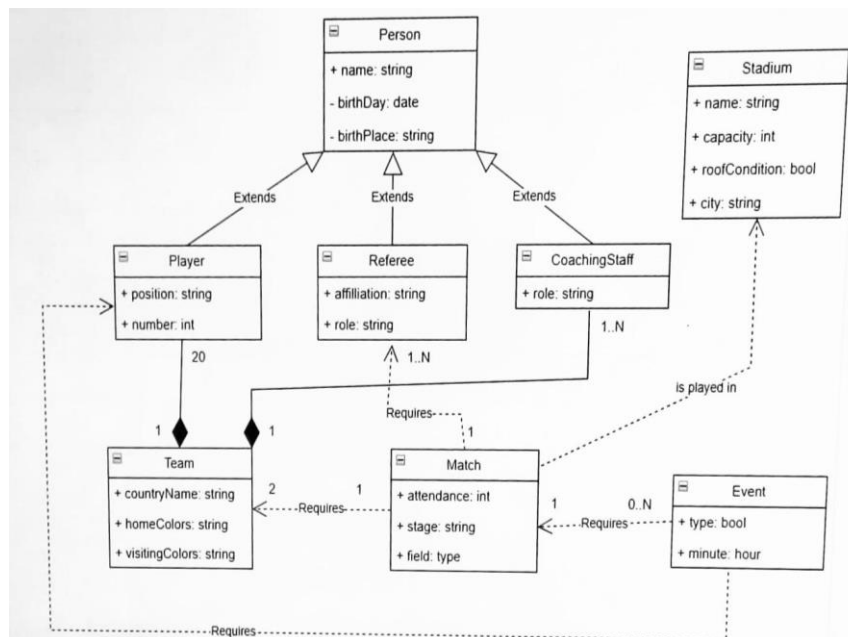
- MakeCalls (Phone Number, Cellphone): This method is used to call a cellphone using its phone number.
- SaveContact (Name, Phone Number): This saves a person's name and phone number in the contact list.
- TakePicture(): This method takes a picture with the phone camera.

2. Use case diagram (online shopping)



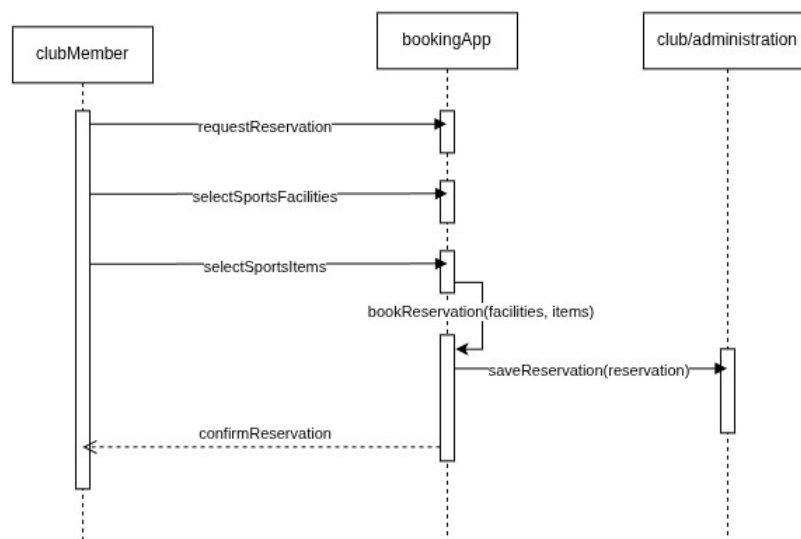
Observations: The diagram shows the main actions involved in an online shopping process. The actors interact with the system to perform the actions shown in the diagram.

3. Class diagram (soccer matches)



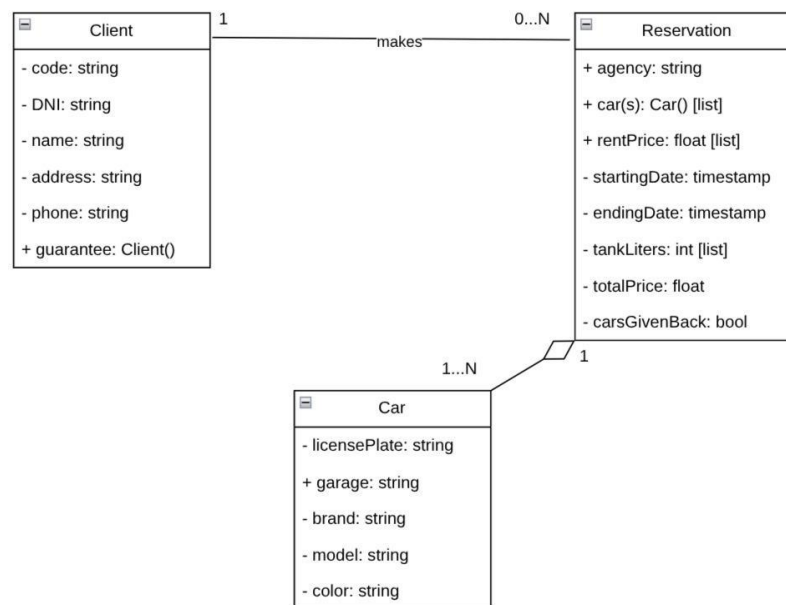
Observations: This diagram represents the different parts of a soccer match and how the classes are connected. The information is organized so it is easier to see the relationship between the elements.

4. Sequence diagram (court rental)



Observations: I think the “clubMember” should be shown as an Actor instead of a class because the person is the one interacting with the process. It would be represented as a person and it would not need an activation box.

5. Class diagram (car reservation)



Observations: In the “guarantee” part, I would remove it from the “Client” class and connect it in the way shown by the diagram so it is clear that “guarantee” is related to “client”.

Also, in the “Reservation” class, I would not leave “Car(s)” as an attribute. It should be connected between the classes to show the relationship more clearly.

Conclusion

To finish, I learned that UML diagrams help us organize an idea before starting to program. They make it easier to see what a system has, what its parts are, and how they connect. For example, class diagrams show the attributes and methods, while use case and sequence diagrams help explain what the user does and how the process happens. I also noticed that small changes in a diagram can make the relationships between the parts much clearer. Overall, these diagrams are useful because they let us understand the project first and then have a better idea of how to build it.